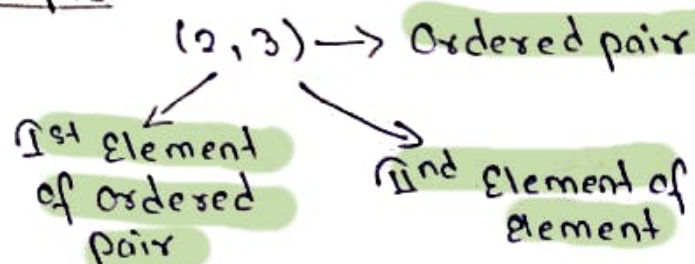


RELATION AND FUNCTION

Ordered Pair

A pair of two objects that appear in a sequence is called an Ordered pair

Example



Note $(2, 3) \neq (3, 2)$

Equality of Two Ordered Pair

Two ordered pairs $[a, b]$ and $[c, d]$ are equal if that corresponding Elements are equal.

that is $[a, b] = [c, d]$
if and only if $a = c, b = d$
 $(a, b) = (c, d)$
 $a = c \text{ \& } b = d$

Q if $(x+3, y-2) = (5, 3)$ then find x & y .

$$x+3=5$$

$$x+5-3$$

$$\boxed{x=2}$$

&

$$y-2=3$$

$$y=3+2$$

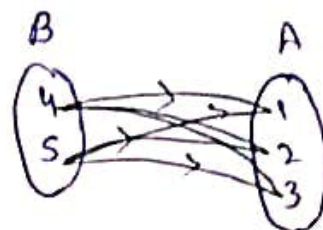
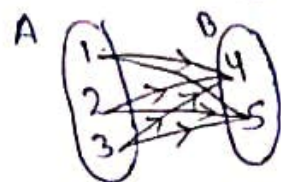
$$\boxed{y=5}$$

Cartesian Product of Sets

The Cartesian product $A \times B$ of any two non-empty Set A and B is the Set of all ordered pairs whose first Element is an Element of the Set A and the Second Element is an Element of the B.

Example

$$n(A) = 3 \rightarrow A = \{1, 2, 3\} \quad n(B) = 2 \rightarrow B = \{4, 5\}$$



$$B \times A = \{(4, 1), (4, 2), (4, 3), (5, 1), (5, 2), (5, 3)\}$$

$$A \times B = \{(1, 4), (1, 5), (2, 4), (2, 5), (3, 4), (3, 5)\}$$

$$n(A \times B) = n(A) \times n(B) = 3 \times 2 = 6$$

$$n(B \times A) = n(B) \times n(A) = 2 \times 3 = 6$$

Note

① $A \times B \neq B \times A$

② $n(A \times B) = n(B \times A) = n(A) \times n(B)$

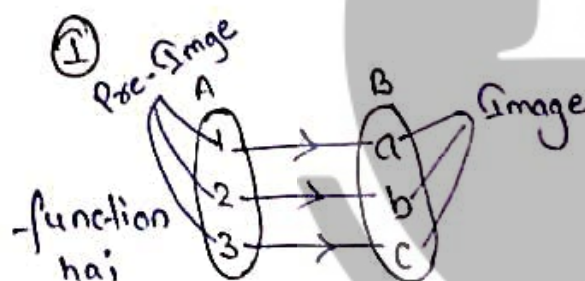
Ex Let $A = \{1, 2, 3\}$ & $n(B) = 5$ then find $n(A \times B) = ?$

$$n(A) = 3$$

$$n(A \times B) = n(A) \times n(B)$$

$$= 3 \times 5$$

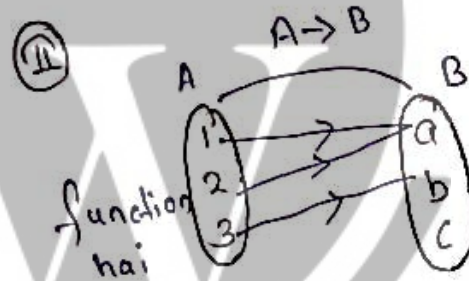
$$= 15$$



$$\text{Domain} = \{1, 2, 3\}$$

$$\text{Range} = \{a, b, c\}$$

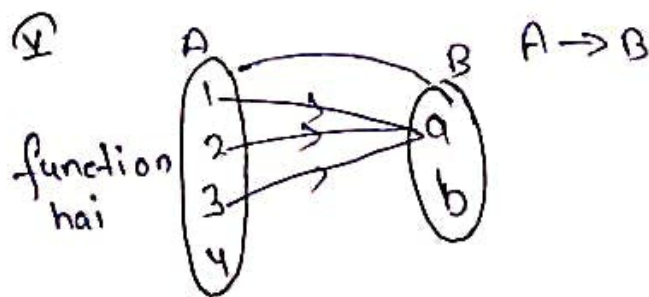
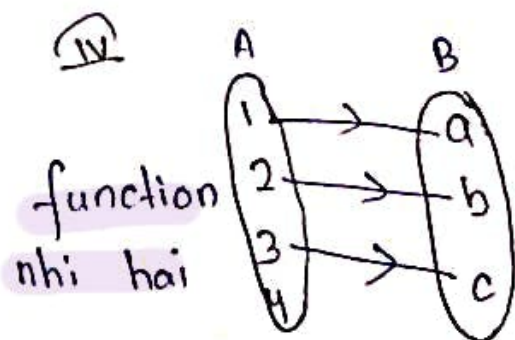
$$\text{codomain} = \{a, b, c\}$$



$$\text{Domain} = \{1, 2, 3\}$$

$$\text{Range} = \{a, b\}$$

$$\text{codomain} = \{a, b, c\}$$



$$\text{Domain} = \{1, 2, 3, 4\}$$

$$\text{Range} = \{a, b\}$$

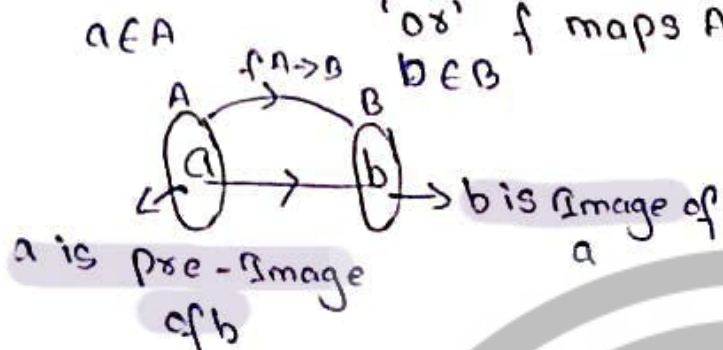
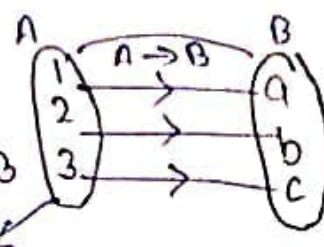
$$\text{codomain} = \{a, b\}$$

Function

A relation R from set A to set B is called a function if each element of A is uniquely associated with some element of B . it is denoted by the symbol:

$$f: A \rightarrow B \quad A \xrightarrow{f} B$$

which means 'f' is a function from A to B
'or' f maps A to B .



1 is pre image of a
2 is pre image of b
3 is pre image of c

a is image of 1
b is image of 2
c is image of 3

Graphical Representation

If a vertical line cuts a given graph at more than one point, then it can not be the graph of a function



Graph (1) represent a function but graph (2) does not represent a function very function is a relation but Every relation is not necessarily a function.

Domain, Co-domain & Range of a Function.

Let $f: A \rightarrow B$ then the set A is known as the domain of f and the set B is known as co-domain f : the set of f image of all the element of A is known as the range of f .

Domain of $f: \{a | a \in A, (a, f(a)) \in f\}$

Range of $f: \{f(a) | a \in A, f(a) \in B\}$

- (a) if Only the rule of function is given then the domain of the functions is the set of those real numbers, where function is defined.
- (b) for a continuous function, the interval from minimum to maximum value of function gives the range. $\rightarrow [0, 1] \rightarrow \text{Range}$
- (c) it should be noted that range is a subset of co-domain.
Range $\subset$ co-domain

Note

- (i) The complete set of all positive real numbers is denoted by R^+ or $R > 0$
- (ii) The complete set of all negative real numbers is denoted by R^- or $R < 0$
- (iii) The complete set of all real numbers other than zero is denoted by R^* or $R \neq 0$
- (iv) The complete set of all integers is denoted by Z

Some Functions and Their Graph

Real function

Domain $\Rightarrow$ Real No. or subset of real no
Range $\Rightarrow$ Real No or subset of Real no

If a Real Funⁿ $f: N \rightarrow N$ such that $f(x) = 2x + 1$

$$f: N \rightarrow N$$

$$y = f(x) = 2x + 1$$

$$x=1 \Rightarrow y = f(1) = 2 \times 1 + 1 = 3$$

$$x=2 \Rightarrow y = f(2) = 2 \times 2 + 1 = 5$$

$$x=3 \Rightarrow y = f(3) = 2 \times 3 + 1 = 7$$

$$x=4 \Rightarrow y = f(4) = 2 \times 4 + 1 = 9$$

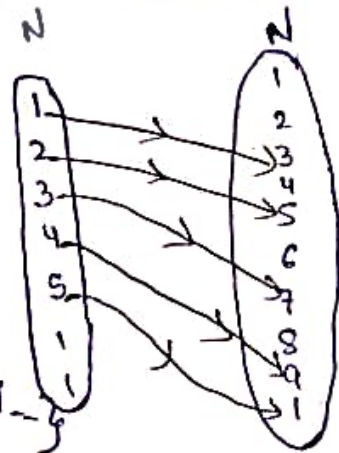
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$$D = N \subset R$$

$$\text{Range} = \{3, 5, 7, 9, \dots\}$$



Some Functions and Their Graph

- > Identity functions
- > Constant functions
- > Modulus functions
- > Signum functions
- > Greatest Integer functions

Identity Functions

The type of funⁿ $y = f(x) = x$ is called Identity function.

$$x, y \in \mathbb{R}$$

$$y = f(x) = x$$

x	-3	-2	-1	0	1	2	3
$y = x$	-3	-2	-1	0	1	2	3

Domain = Real number = $\mathbb{R}$

Range = Real number = $\mathbb{R}$

Constant Functions

$y = f(x) = c$ is called a constant function.

Ex

$$y = f(x) = 2 \quad f(x) = 3$$

$$f(x) = -4 \quad f(x) = -5$$

$$y = f(x) = c$$

x	-3	-2	-1	0	1	2	3
$y = f(x) = c$	c	c	c	c	c	c	c

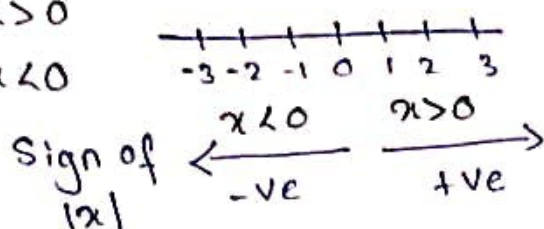
Domain = Real Number

Range = $\{c\}$

Modulus Functions

$y = |x|$ is called Modulus function.

$$y = \begin{cases} x & \text{when } x > 0 \\ -x & \text{when } x < 0 \end{cases}$$



x	-3	-2	-1	0	1	2	3
$y = x $	3	2	1	0	1	2	3

Domain = Real numbers
Range = $[0, \infty)$

Signum Functions

A function $f(x) = \begin{cases} \frac{|x|}{x} & \text{when } x \neq 0 \\ 0 & \text{when } x = 0 \end{cases}$ is called a signum function.

$$y = f(x) = \begin{cases} \frac{|x|}{x} & \text{when } x \neq 0 \\ 0 & \text{when } x = 0 \end{cases}$$

$$y = f(x) = \begin{cases} \frac{x}{x} & \text{when } x > 0 \\ -\frac{x}{x} & \text{when } x < 0 \\ 0 & \text{when } x = 0 \end{cases}$$

$$|x| = \begin{cases} x & \text{when } x > 0 \\ -x & \text{when } x < 0 \end{cases}$$

$$y = f(x) = \begin{cases} 1 & \text{when } x > 0 \\ -1 & \text{when } x < 0 \\ 0 & \text{when } x = 0 \end{cases}$$

$$y = f(x) = \begin{cases} 1 & \text{when } x > 0 \\ -1 & \text{when } x < 0 \\ 0 & \text{when } x = 0 \end{cases}$$

$$\begin{aligned} y = f(x) &= 1 & \text{when } x > 0 \\ y = f(x) &= -1 & \text{when } x < 0 \\ y = f(x) &= 0 & \text{when } x = 0 \end{aligned}$$

Domain = $\mathbb{R}$

Codomain = $\mathbb{R}$

Range = $\{-1, 0, 1\}$

Greatest Integer Functions

A function $y = f(x) = [x]$ is called a greatest Integer function and its value equal to x or less than x .

$$y = [x]$$

$$[0.0] = 0$$

$$[0.1] = 0$$

$$[0.1) = 0 \quad [0.2] = 0$$

$$[0.2) = 1 \quad [0.3] = 0$$

$$[2,3) = 2 \quad ", 1$$

$$[3,4) = 3 \quad [0.9999) = 0$$

$$[1.0] = 1$$

$$[1.1] = 1$$

$$[1.2] = 1$$

$$[1.3] = 1$$

$$1, \therefore$$

$$[1.9999] = 1$$

$$[2.0] = 2$$

$$[2.1] = 2$$

$$[2.2] = 2$$

$$1, \therefore$$

$$[2.9999] = 2$$

$$[3.0] = 3$$

$$[3.1] = 3$$

$$[3.2] = 3$$

$$", \therefore$$

$$[3.99] = 3$$

$$y$$

$$[0,1) = 0$$

$$[1,2) = 1$$

$$[2,3) = 2$$

$$[3,4) = 3$$

$$y$$

$$[-1,0) = -1$$

$$[-2,-1) = -2$$

$$[-3,-2) = -3$$